

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is instructed. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate the performance based on evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

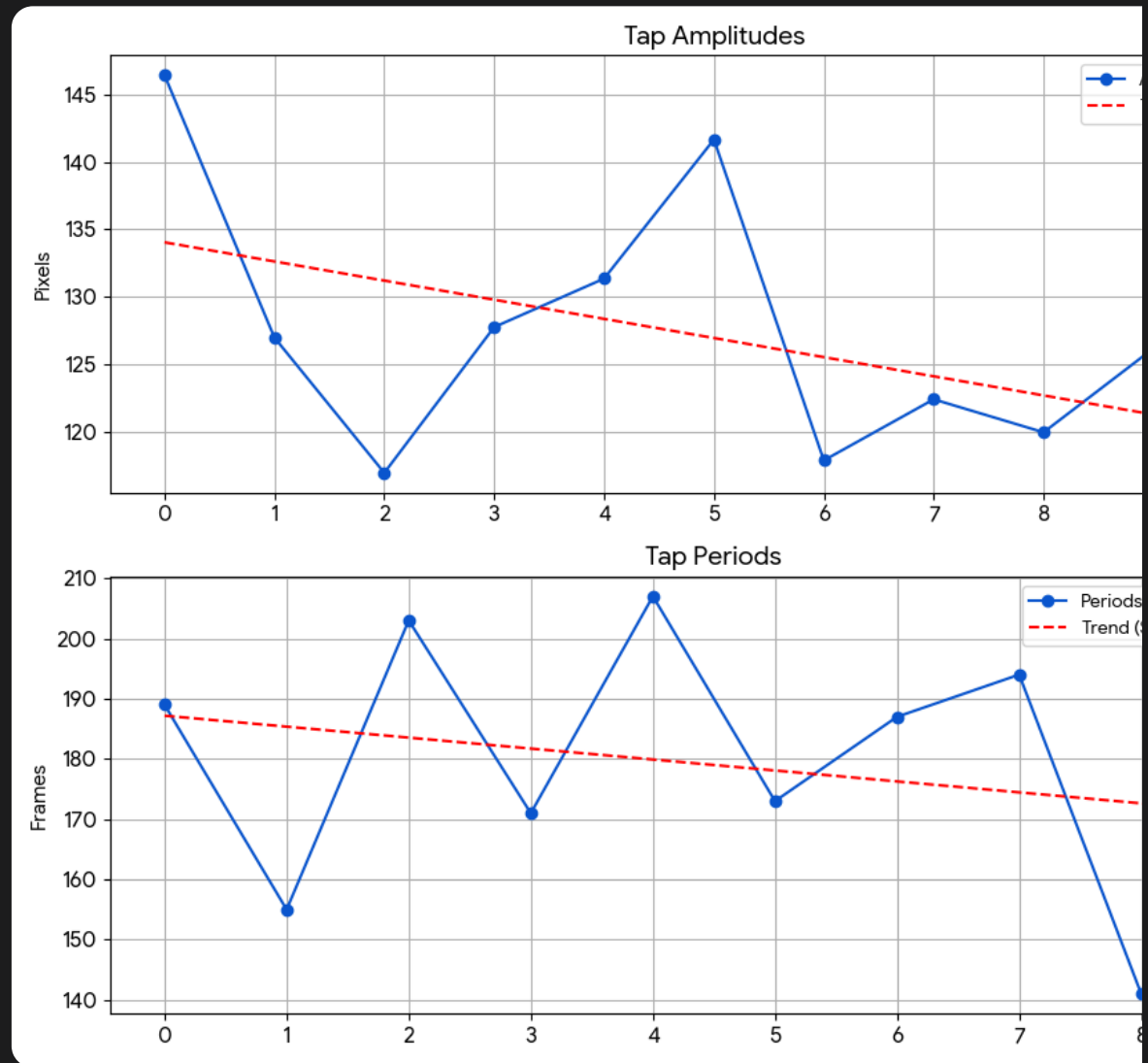
Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide a score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]

Tap amplitudes (pixels): [146.40013661195812, 127.00393694685216, 117.00427342623003, 127.781062759131.36590120727678, 141.65097952361643, 117.92370414806346, 122.44182292011173, 120.0, 126.21014222118.92854997854805]



Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 scoring criteria:

1. Amplitude Analysis:

- **Data:** The amplitudes (in pixels) for the 11 taps are: [146, 127, 117, 128, 131, 142, 118, 122, 120, 126, 119].
- **Trend:** The first tap is the largest (146). There is an immediate and significant drop in amplitude by the second tap (117).
- **Recovery:** Although there is a partial recovery around the 6th tap (142), the amplitude drops significantly again at the end of the sequence.
- **Criteria Match:** The immediate reduction in amplitude aligns with the criterion: "The amplitude decrements start at the first tap." This is a specific indicator for **Score 3 (Moderate)**.

2. Rhythm and Speed Analysis (Periods):

- **Data:** The periods (intervals between taps) are: [189, 155, 203, 171, 207, 173, 187, 194, 141, 170].
- **Trend:** The rhythm is highly irregular, oscillating between fast (~140-155) and slow (~190-207) intervals throughout the sequence.
- **Interruptions:** If we consider the faster taps as the baseline capability, the longer intervals (189, 203, 207, 187, 194) represent hesitations or interruptions in the rhythm. There are approximately 5 such interruptions/irregularities in the 10-interval sequence.
- **Criteria Match:** The presence of frequent interruptions (3 to 5 or more) suggests a score of **2 (Mild)** or **3 (Moderate)**.

3. Speed (Slowing):

- **Trend:** Linear regression on the periods shows a negative slope (getting faster/shorter intervals on average), so pr is not the primary deficit here. The issue is variability (hesitations) and amplitude decrement.

Conclusion: The most severe feature determines the score. The subject exhibits an amplitude decrement that begins in first tap (dropping from 146 to 127/117), which matches the criteria for a **Moderate** impairment. Additionally, the rhythm is disrupted with multiple hesitations.